

Figure 1: Bash variables to customise the prompt



Figure 2: More customised *bash* prompts

number of demo characters and then press *Enter*, to see a continuation prompt (normally >) till you type the closing double quotes ("). You can change *PS2* to show something different, like we did for PS1.

PS3 and *PS4* customise the shell output in case of shell scripts; PS3 sets the prompt for the select command in shell scripts, and PS4 sets the value printed after the PS1 prompt when you enable script execution tracing using *set -x*.

Note: Source files accompanying this article are available for download from the LFY website at http://linuxforu.com/article_source_code/aug11/console.zip

Bash executes the command in *PROMPT_COMMAND* every time (before) it shows the PS1 prompt. You could use this to pre-process or show some information before the prompt. To see the things we've discussed in action, run the accompanying *ps1tocommand.sh* with so*urce ps1tocommand.sh*, or *. ps1tocommand.sh* (the dot is a short-cut for the source command). You have to source the script into your current shell because if you run it in a new shell (e.g., *sh ps1tocommand.sh*), the settings are changed only in that shell, are lost when the script finishes running, and do not affect the prompt displayed in the (parent) shell in your terminal. After sourcing this script, to reset to the previous prompt settings, run *PROMPT_COMMAND=$PCVAL; PS1=$PS1VAL*.

The output of this is shown in Figure 1. If you noticed, the original value of *PS1* contains code like *\u*, *\w*, *\$* etc. As per its man page, *bash* provides the backslash-escaped special characters in Table 1 to customise prompt strings.

\a	an ASCII bell character
\d	date in "Weekday Month Date" format
\h	hostname up to the first '.'
\H	hostname
\j	number of jobs currently managed by the shell
\l	basename of the shell's terminal device name
\s	basename of the shell
\t	current time in 24-hour HH:MM:SS format
\T	current time in 12-hour HH:MM:SS format
\@	current time in 12-hour am/pm format
\A	current time in 24-hour HH:MM format
\u	username of the current user
\v	version of *bash*
\V	version + patch level of *bash*
\w	current working directory
\W	basename of the current working directory
\!	history number of the command
\#	command number of the command
\$	a # for super user, otherwise a $

Table 1: PS1 special characters

You can use any quick-running command in prompt variables or scripts, besides these special characters. ('Quick-running', since the shell has to interpret these every time it displays the prompt.) Also, if you want to put a lot of text in your prompt, it's better to divide the content between *PROMPT_COMMAND* and *PS1*. Source into your *bash* shell the accompanying *spiceyprompts.sh*, then *run prompt1, prompt2, prompt3* to see some more prompts that combine various shell commands, escape characters, etc. Finally, run *rov* to restore the original prompt.

Figure 2 shows the output after doing this. All the examples used till now only affect the shell instance in which they are typed or invoked. To make settings 'permanent' after testing them out, add the commands to your *~/.bashrc* profile.

ANSI escape sequences and *bash* glorification

Now what about using colours in the shell? Almost all consoles, whether physical or terminal emulators, support various character combinations that control text formatting, colour and other aspects. These are known as ANSI escape sequences. We see these used in console-mode GUI applications like installers, text dialogue utilities, and tool-kits like *ncurses*, *newt*, etc. This section is a hands-on introduction to ANSI escape sequences to colourise the console. We will explore *ncurses* and *newt* in the second part of the series.

These escape sequences control various aspects of the text to be displayed, and also the text cursor and graphics modes. The most commonly used format of the terminal escape sequence is *\033[[text colour;][text background;][text attribute]m[text]\033[m*. A listing of the most common numbers to choose the text colour, background and attribute, is shown in Table 2. You can change only the setting you're